

AFTERWORD

FACTOR MANAGEMENT

The first model of factor investing, the capital asset pricing model, was developed in the 1960s and suggested that one factor, the market portfolio, determined asset risk premiums. But at the time, investors had no way to invest in the market factor. The market index fund was created a decade later and gave investors a way to collect the equity market risk premium through low-cost exposure to the whole equity market. Market index funds remained niche products until the 1980s and really took off in the 1990s; now many investors invest in the market factor cheaply, in scale, and with minimal agency issues. It took twenty to thirty years, however, after the conception of the theory for investing via the market portfolio factor to become widespread.

The theory behind moving beyond just the market factor to a world of multiple factors was developed in the 1970s. Theoretical developments and substantial empirical work validating multiple sources of factor risks occurred during the 1980s and 1990s. Today we are experiencing exciting times in factor investing as industry innovates to create low-cost, scalable versions of factors that are diversified within and across asset classes and regions. A new generation of dirt-cheap factor index funds will allow investors to harvest many sources of risk premiums, including those associated with value versus growth, momentum investing, credit risk, and volatility risk. Most of the effort has been concentrated in dynamic factors that take time-varying nonmarket capitalization weighted positions, but there are also efforts to go further by linking the underlying macro factors to asset allocations.

I see three main areas where we still fall short of the theory. These are also areas where industry can help us reach the full potential of factor investing:

Identifying and Managing Bad Times

This book started with the asset owner. Her fears, desires, the riskiness of her income, her liabilities, how she perceives different sources of risk—all uniquely